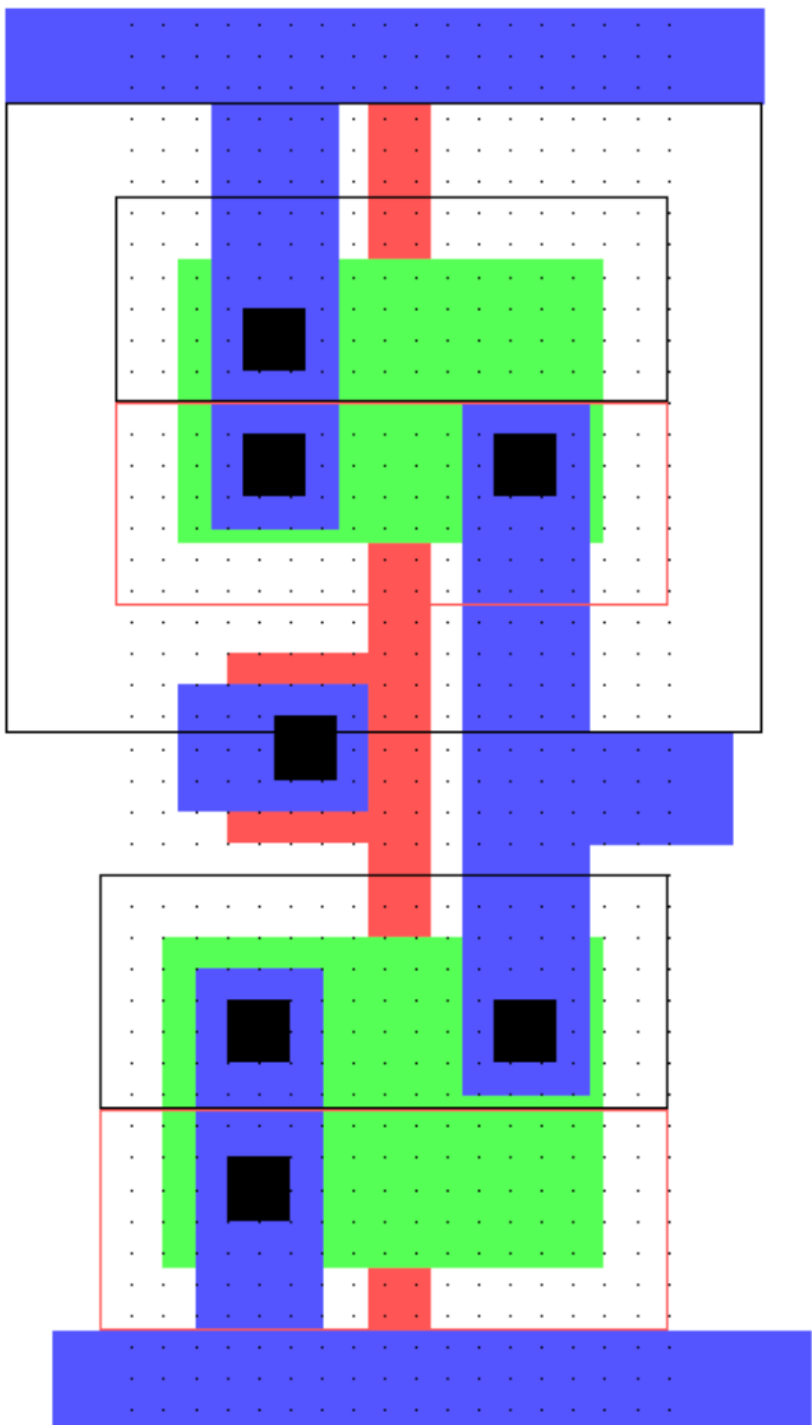
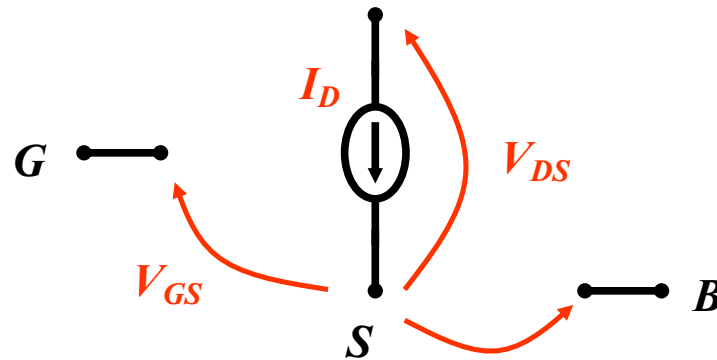




Il MOS

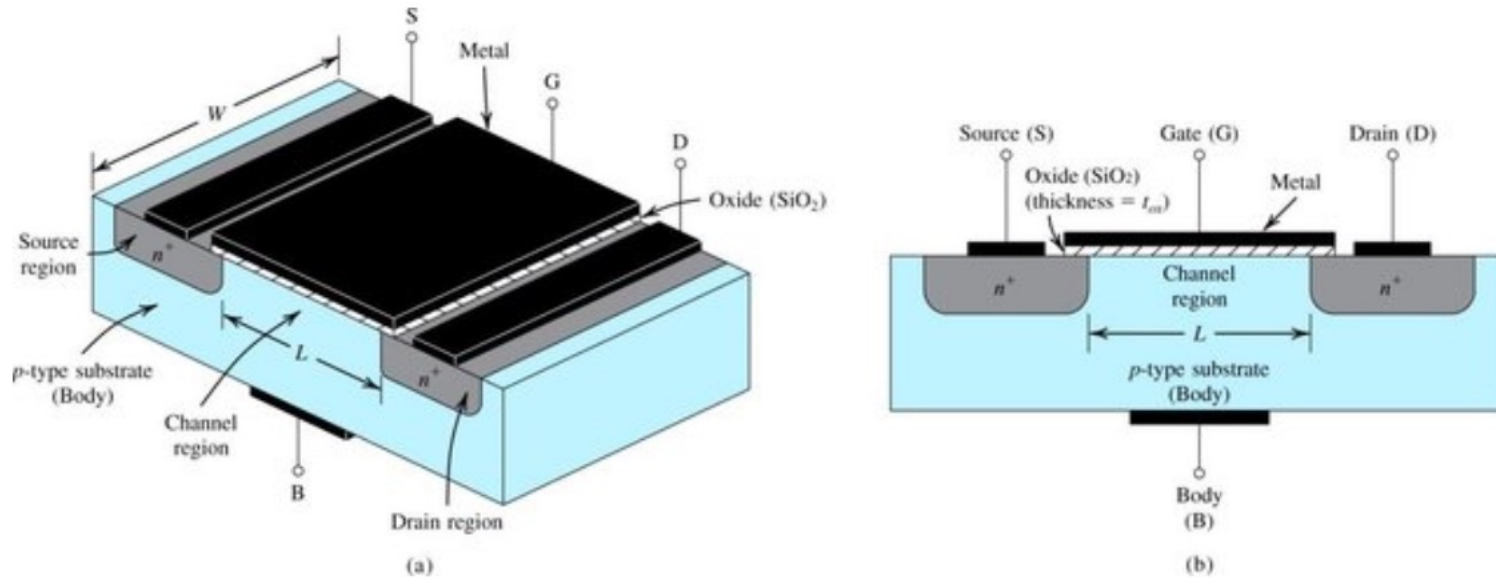
- Il MOS è un dispositivo non lineare a 4 terminali
- Lo vediamo come un generatore di corrente controllato in tensione:

$$I_D = f(V_G, V_S, V_D, V_B) = f(V_{GS}, V_{DS}, V_{BS})$$



- MOS già visto in altri corsi: Fondamenti di Elettronica Lineare, Dispositivi, Microelettronica
- MOS dal punto di vista del progettista di circuiti integrati
- L'attenzione si concentra su:
 - modello per grandi segnali,
 - modello per piccoli segnali bassa freq.,
 - modello per piccoli segnali alta freq.

- Modello per grandi segnali: $I_D = f(V_{GS}, V_{DS}, V_{BS})$;
- Modello per piccoli segnali in bassa frequenza:
Transconduttanza g_m
Conduttanza di uscita g_o ;
- Modello per piccoli segnali in alta frequenza:
Effetti parassiti capacitivi



La condizione di lavoro che ci interessa è l'inversione (formiamo il canale): debole (sottosoglia), e forte (soprasoglia, lineare e saturazione)

...E come progettisti ci interessa l'overdrive, o ΔV , o $V_{GS} - V_{th}$

Overdrive, W/L , L e la scelta di schemi circuitali furbi sono la «console» di chi progetta

- Esistono due livelli di inversione che dipendono solo dall'overdrive: debole e forte
- Debole (sottosoglia): $0 < V_{GS} - V_{TH} < 4KT/q$ ($\sim 100mV$): canale non completamente formato, comportamento simile al bipolare
- Forte: $V_{GS} - V_{TH} > 8KT/q$ ($\sim 200mV$): canale completamente formato

Anche la tensione V_{DS} è importante, perché ci dice se il MOS è in lineare (si comporta come un resistore) o è in saturazione

Spesso la V_{DS} dipende dal carico o da altri transistor presenti nel circuito complessivo

Se $V_{DS} > V_{GS} - V_{Th}$ il MOS è in saturazione, se no è in lineare

...vediamo ora l'equazione della corrente nelle varie condizioni operative...

Debole inversione

$$I_D = I_{D0} e^{\frac{V_G}{nkT/q}} \left(e^{-\frac{V_S}{nkT/q}} - e^{-\frac{V_D}{nkT/q}} \right) \quad I_{D0} = \mu C_{ox} \frac{W}{L} e^{-\frac{V_{th}}{nkT/q}}$$

$$I_D \approx I_{D0} e^{\left(\frac{V_{GS}}{nkT/q} \right)} \quad \text{In saturazione}$$

$$I_D \approx 0 \quad \text{In lineare}$$

Inversione

$$I_D = \mu C_{ox} \frac{W}{L} \left(V_{GS} - V_{th} - \frac{V_{DS}}{2} \right) V_{DS}$$

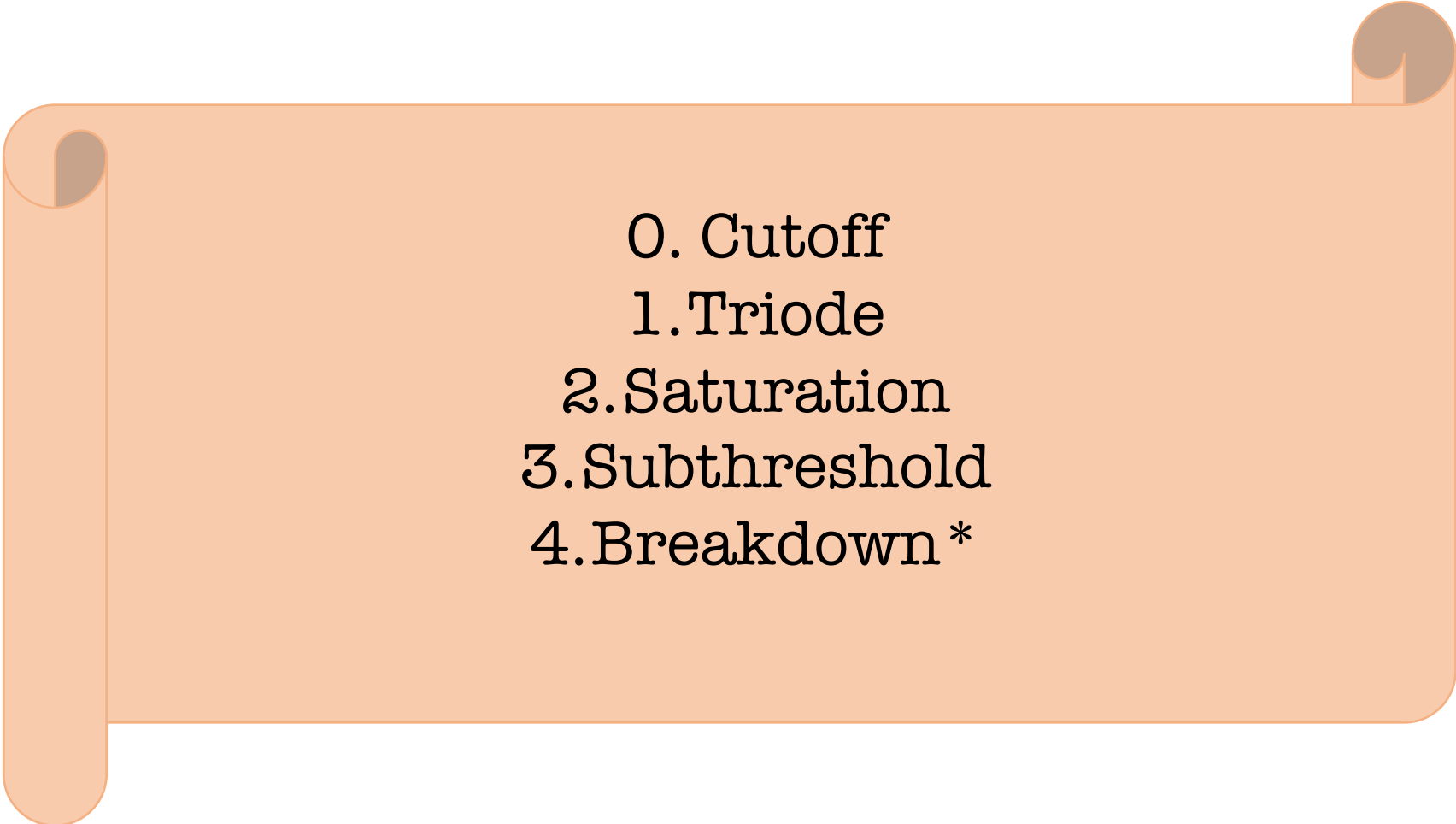
In lineare

$$I_D = \frac{1}{2} \mu_0 C_{ox} \frac{W}{L} \frac{(V_{GS} - V_{th})^2}{1 + \left(\frac{\mu_0}{2v_{sat}L} + \theta \right) (V_{GS} - V_{th})}$$


In saturazione

Effetti del 2° ordine: saturazione velocità e deterioramento mobilità per c.e. verticale

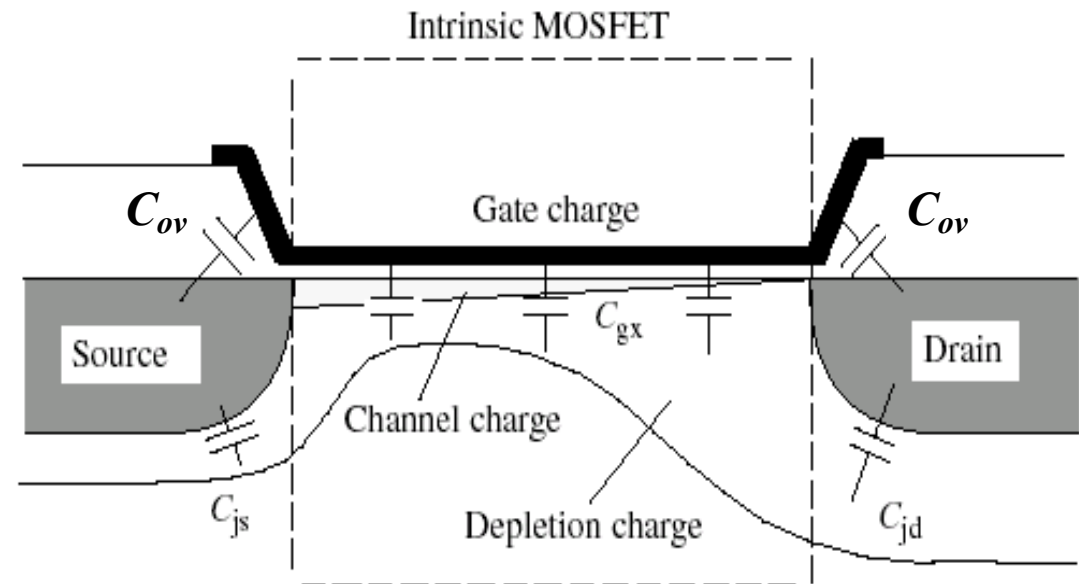
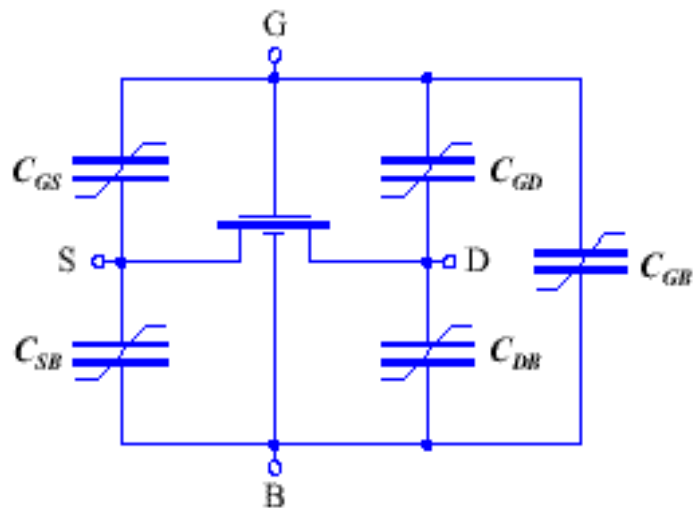
Mappa regioni operative in Cadence

- 
- A large orange scroll graphic with rounded corners and a vertical strip on the left side, containing a list of operating regions.
- 0. Cutoff
 - 1. Triode
 - 2. Saturation
 - 3. Subthreshold
 - 4. Breakdown *

* La simulazione non mostra che il dispositivo si è rotto, avvisa solo

Comportamento in frequenza: le capacità parassite

Dipendono dalla regione operativa, dal punto di lavoro e dalle dimensioni



Il modello BSIM3 e le capacità parassite

Node	Gate	Drain	Source	Bulk
	+cggb			-cggb
Gate	+capgdo+cgdb	+cgsb		-cgdb
	+capgso	-capgdo	-capgso	-cgsb
	+capgbo			-capgbo
		+cddb		-cdgb
Drain	+cdgb	+capgdo+cdsb		-cddb
	-capgdo	+capbd		-cdsb
				-capbd
	-cggb	-cgdb	-(cgsb+cdsb+cbsb)	
Source	-cdgb	-cddb	+capgso	(*)
	-cbgb	-cbdb	+capbs	-capbs
	-capgso			
	+cbgb	+cbdb	+cbsb	-(cbgb+cbdb+cbsb)
Bulk	-capgbo	-capbd	-capbs	+capgbo
				+capbd+capbs

Il modello BSIM3 presenta un sacco di sigle di capacità parassite e alcune hanno valore negativo, dovuto alla rappresentazione matriciale (non ha un senso fisico).

Capacità in debole inversione, saturazione

$$C_{GS} = C_{GD} \cong WC_{ov}$$

$$C_{GB} = WLC_{ox}$$

$$C_{SB} = C_{DB} = f(V_{SB}, V_{DB})$$

Gm in debole inversione, saturazione

$$g_m = \frac{I_D}{kT / q} = \frac{I_D}{26mV}$$

Capacità in inversione, lineare

$$C_{GS} = C_{GD} \cong WL / 2 \cdot C_{ox} \qquad C_{GB} \approx 0$$

$$C_{SB} = C_{DB} = f(V_{SB}, V_{DB})$$

Gm e Go in inversione, lineare

$$g_m = \mu C_{ox} \frac{W}{L} V_{DS}$$

$$g_o = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th} - V_{DS}) \approx \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})$$

Capacità in inversione, saturazione

$$C_{GS} = 2/3 WLC_{ox} + WC_{ov}$$

$$C_{GD} = WC_{ov}$$

$$C_{SB} = C_{DB} = f(V_{SB}, V_{DB})$$

$$C_{GB} \approx 0$$

Gm e Go in inversione, saturazione

$$g_m = \frac{2I_D}{V_{GS} - V_{th}} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th}) = \sqrt{\mu C_{ox} \frac{W}{L} I_D}$$

$$g_o \propto (I_D, L^{-1})$$

Short channel

$$g_o \propto (\sqrt{I_D}, L^{-1}) \propto (g_m, L^{-1})$$

Long channel

Ripassiamo i parametri principali nelle varie condizioni operative

Transconductance

- Subthreshold region (like a bipolar transistor):

$$g_m = -g_{mb} = \frac{I_D}{n \frac{kT}{q}}$$

- Linear region:

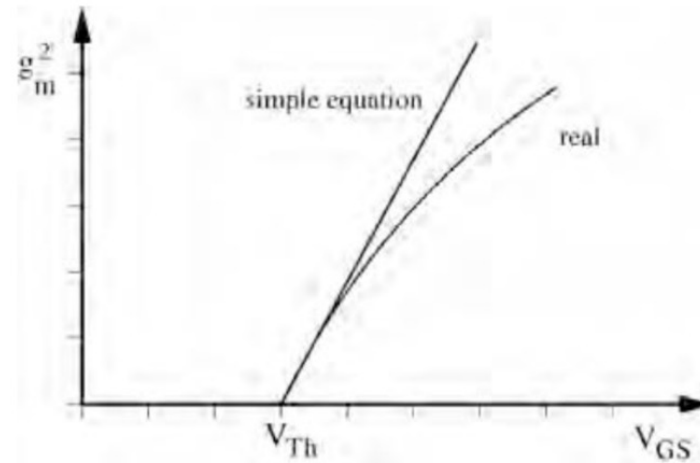
$$g_m = \mu C_{ox} \frac{W}{L} V_{DS}$$

- Saturation region:

$$g_m = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{Th}) = \frac{2I_D}{V_{GS} - V_{Th}} = \sqrt{2\mu C_{ox} \frac{W}{L} I_D}$$

In the real situation g_m is smaller than the value predicted by the given simple equation.

Moreover g_m changes because of the dependence of μ from the temperature, the transversal and the mean lateral electric field.



$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{-\frac{3}{2}} \left(1 + \frac{E_y}{E_{\text{crit}}} \right)^{-m} \frac{1}{1 + \frac{V_{DS}}{LE_{\text{sat}}}}$$

SPICE uses the fitting equation:

$$\mu = \mu_0 \left(\frac{U_{\text{crit}} \epsilon_{\text{Si}}}{C_{\text{ox}} (V_{\text{GS}} - V_{\text{on}} - U_{\text{tra}} - V_{\text{DS}})} \right)^{U_{\text{exp}}}$$

Drain output conductance (g_{ds})

- Linear region:

$$g_{ds} = \mu C_{ox} \frac{W}{L} (V_{GS} - V_{Th} - V_{DS})$$

- Saturation region (first order):

$$g_{ds} = \lambda I_D$$

Second order effects

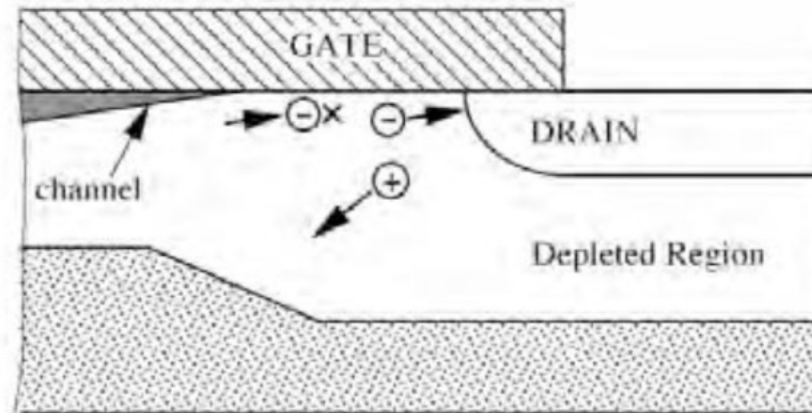
- The channel length reduction has been calculated taking into account only the lateral drop voltage. A more accurate analysis gives:

$$\frac{1}{\Delta L} \cong \frac{1}{\Delta L_{(1^{st} \text{ order})}} + \frac{C_{ox}}{\epsilon_S} \left[\frac{\alpha(V_{DS} - V_{GS}) + \beta(V_{GS} - V_{sat})}{V_{DS} - V_{sat}} \right]$$

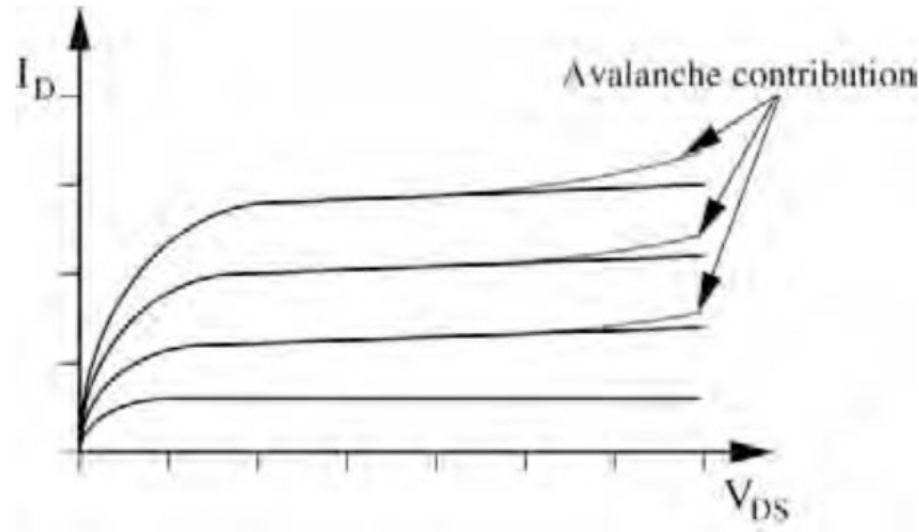
- The threshold voltage depends on the V_{DS} (short channel)
- The mobility depends on the lateral mean field V_{DS} / L
- The avalanche effect increases I_D

Avalanche current

Mobile charges, accelerated by electric field in the drain depleted region, creates, by impact ionization, electron-holes pair.

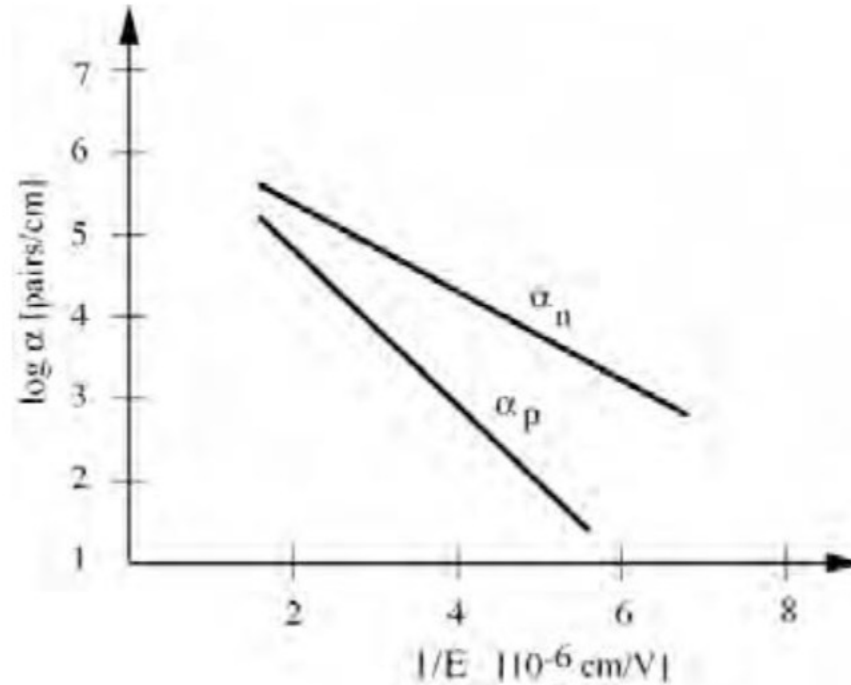


The drain current I_D and a substrate current I_B increase due to this contribution.



- The substrate current may contribute to latch-up
- The device noise increases
- The output impedance decreases
- Carriers can be trapped on the oxide and V_{Th} changes (hot electron effect)

- Avalanche current worse in n-channel



More accurate expression of the output conductance:

$$g_{ds} = \lambda I_D - g_m \frac{\partial V_{Th}}{\partial V_{DS}} + \frac{I_D}{\mu} \frac{\partial \mu}{\partial V_{DS}} + \frac{\partial I_S}{\partial V_{DS}}$$

(first order) (short channel) (velocity saturation) (avalanching)

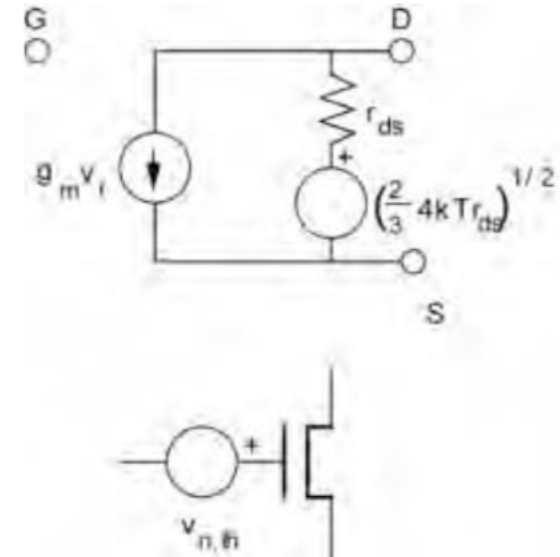
Noise

Thermal noise

Due to the finite output resistance:

$$\overline{V_{nth}^2} = 4\gamma kT \frac{1}{g_m}$$

$\gamma = 2/3$, for $I_D = 50 \mu A$, $(V_{GS} - V_{Th}) = 300 \text{ mV}$, $V_{nth} = 5.6 \text{ nV}/\sqrt{\text{Hz}}$



If the bandwidth BW of the system is $f_2 - f_1$, the input referred noise voltage is:

$$\overline{V_n^2} = \int_{f_1}^{f_2} \frac{\overline{V_{nth}^2}}{\Delta f} df = 4 \frac{2}{3} kT \frac{1}{g_m} \text{BW}$$

Flicker noise

Due to the trapping and detrapping of carriers by surface states at different energy levels.

Modeled as:

$$\frac{\overline{i_{nf}^2}}{\Delta f} = \frac{2 K_f I_D}{C_{ox}^{k_c} L^2 f^\alpha}$$

$$\frac{\overline{v_{nf}^2}}{\Delta f} = \frac{\overline{i_{nf}^2}}{\Delta f g_m^2} = \frac{2 K_f I_D}{C_{ox}^{k_c} L^2 f^\alpha g_m^2} = \frac{K_f}{\mu C_{ox}^{k_c+1} W L f^\alpha}$$

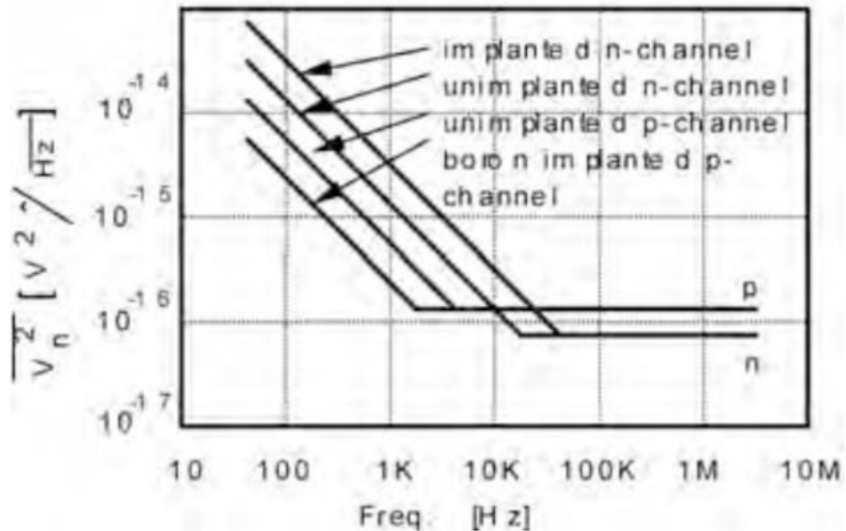
Typically $\alpha \cong 1$, $k_f \cong 1$, $k_c + 1 \cong 1$,

$V_{nf} = 40 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz with $W \cdot L = 1000 \text{ } \mu\text{m}^2$.

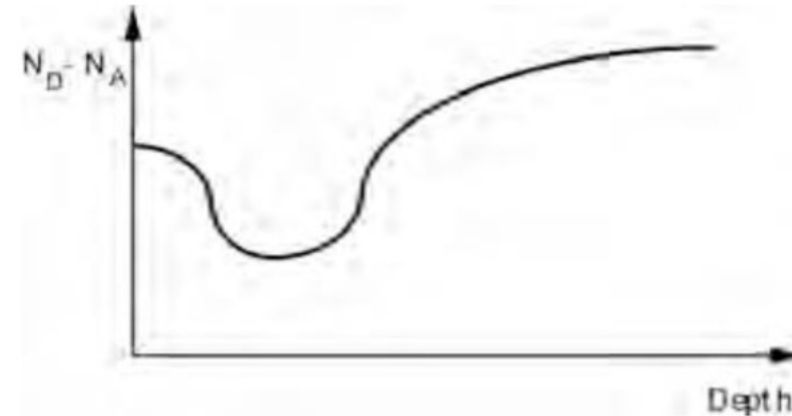
The power of the flicker noise is concentrated at low frequency.

$$\overline{V_{nf}^2} = \int_{f_1}^{f_2} \frac{\overline{V_{nf}^2}}{\Delta f} df = \frac{K_f}{\mu C_{ox} W L} \ln\left(\frac{f_2}{f_1}\right)$$

Noise spectra for n-channel and p-channel transistors.



Boron implanted p-MOS has low $1/f$ noise (buried channel)



Avalanche noise

Due to the statistical fluctuation in the number of carriers of the avalanche current (shot noise). Modeled as:

$$\overline{\frac{i_{nav}^2}{\Delta f}} = 2 q I_{av} \quad \overline{\frac{v_{nav}^2}{\Delta f}} = \frac{\overline{i_{nav}^2}}{\Delta f g_m^2} = \frac{q I_{av}}{\mu C_{ox} \frac{W}{L} I_D}$$

If we compare thermal noise and avalanche noise, we have:

$$\frac{v_{nav}^2}{v_{nth}^2} = \frac{q I_{av}}{\mu C_{ox} \frac{W}{L} I_D} \frac{g_m}{4 \gamma k T} = \frac{V_{GS} - V_{Th}}{4 \gamma \frac{k T}{q}} \frac{I_{av}}{I_D}$$

we get comparable noise if I_D / I_{av} is of the same order of $(V_{GS} - V_{Th}) / (kT/q)$. The avalanche current at $V_{DS} = 5$ V can be of the order of $0.5 \div 1 \mu\text{A}$.

To minimize noise

- Thermal noise:
 - use large g_m (large W / L)
 - use low series resistance (connection and gate resistance)
- Flicker noise:
 - use large device area ($W \cdot L$)
 - use thin oxide (high C_{ox})
 - use "clean" technology (low N_{ss})
 - try to get buried channel
 - use p-channel devices
- Avalanche noise:
 - reduce V_{DS}
 - use p-channel devices